Assignment - 1
DBMS

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Section - A.

Q1. What is DBMS ? Why is it important ?

→ A DBMS (Database Management System) is software that stores, manages and retrieves data in an organised way It is required to reduce data redundancy, maintain consistency. Provide security, and allow easy data access.

Q2. Why is DBMS better than the traditional file system ?

→ DBMS reduces duplication of data, avoids inconsistency provides security, supports multiple users at the same time and offers backup and recovery facilities.

Q3. What is Data Abstraction ? Explain its levels.

→ Data Abstraction means hiding internal details and showing only necessary information.

Three levels are :

- Physical level - How data is stored.
- Logical level - What data is stored and relationship.
- View level - How use see the data.

Q4. What is Three - Schema Architecture ?

→ Three - schema architecture includes internal, conceptual and external levels. it is important because it provides data independence and seperates users views from physical storage.

Q5. What is data independence ? Why is it required ?

→ Data Independence means changes at one level of database don't affect other levels. It is required for flexibility and easy modification of the system.

Q6. What is a ER Model ?

Ans → ER Model is conceptual diagram that represents entities, their attributes and relationship between them in a graphical form.

Q7. What are integrity constraints?

→ Integrity constraints are rules that maintain accuracy and consistency of data. Examples are privacy key, foreign key and unique and NOT NULL constraints.

Section B: Design. Draw & Defend.

(i) Three - schema

External level → user views

↓

conceptual level → overall structure

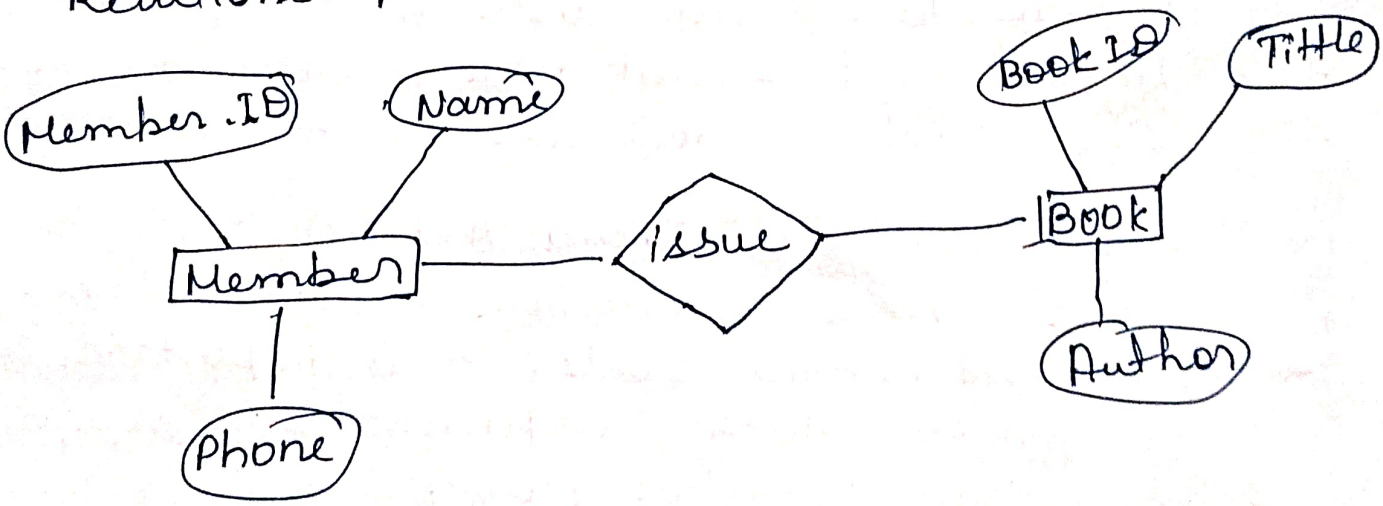
↓

Internal level → physical storage.

(ii) Library ER Diagram

- Book (Book ID, Tittle, Author)
- Member (Member ID, Name, Phone)

Relationship: Member issue Book.

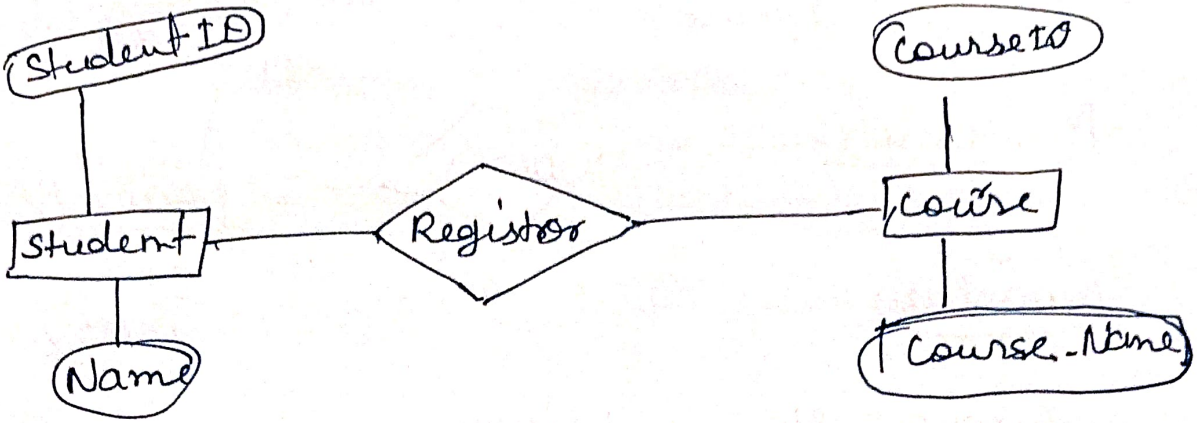


(iii) Student - course Registration ER:

- Student (Student ID, Name)
- course (course ID, course, Name)

Relationship: Student Registration

Entity Relation (ER).



Section C : Case Based Analytical Questions.

(i) ER - Diagram - Hospital System.

Entities : Patient, Doctor, Appointment

Patients book Appointment

Doctor attends Appointment

Primary keys : Patient_ID, Doctor_ID, Appointment_ID

(ii) Online shopping :

Customer (customer_ID - PK)

Product (Product_ID - PK)

Order (Order_ID - PK)

Foreign key in order Table connect customer and Product.

(iii) University - Data Abstraction :

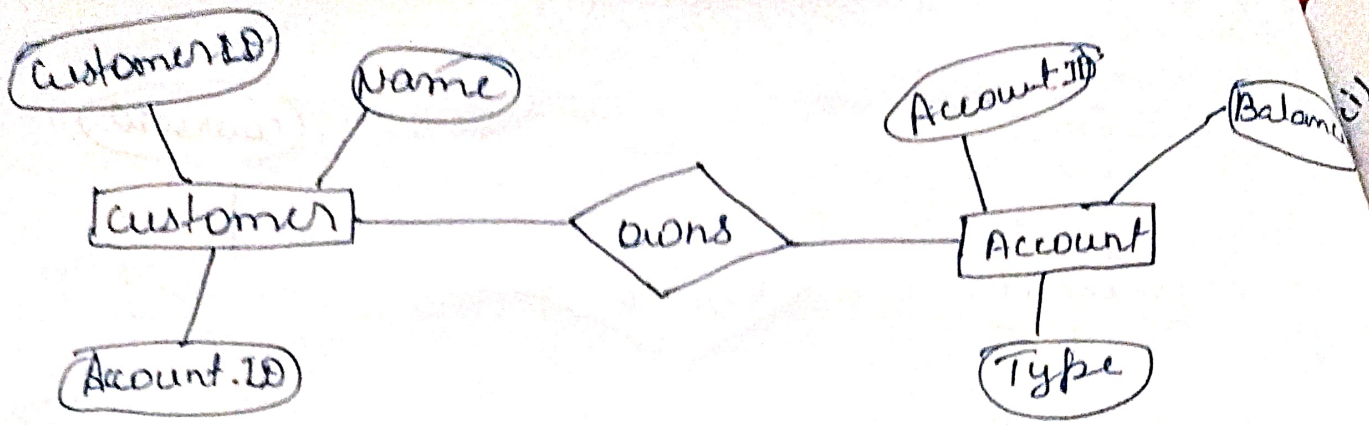
Student see marks (view level)

Admin manages records (logical level)

DBA controls storage (Physical level)

Section D : One Diagram + one INSight.

(i) ER - diagram - ATM system



Account_No is Primary key
Relation : customer owns Account

Account_No take uniquely identifies Account

(ii) Three-Schema Architecture + one benefits

Benefit : Provides Data Independence - changes storage don't affect users.

Diagram : External level (user view)
↓
conceptual level (logical structure)
↓
Internal level (Physical storage level)

(iii) Draw a Entity with attributes

Entity : Employee
Employee_ID (PK)
Name
~~Stan~~ Salary
Department

Primary key : Employee_ID
because for all employee their ID will be different.

Section - 5 : Reflection, Assumption & viva

Two assumption while drawing ER diagram

- Each student has a unique student ID
- One appointment is linked to only one doctor.

(ii) One limit of ER - Modeling :

- Cannot represent complex constraint easily.
- Doesn't show procedural aspects
- Not suitable for large dynamic system directly.

(iii) Most Difficulty Part in Unit 1.

The most difficult was understanding data independence, because it requires clear understanding of three-schema architecture and how levels interact.